

Experiment No. 17

Title

Prolog Program to Sum the Integers from 1 to n

Aim

To write a Prolog program to find the sum of integers from **1 to n**.

Objective

The objective of this experiment is to understand recursion in Prolog by calculating the sum of natural numbers from 1 to a given number **n**.

Algorithm

Step 1: Start.

Step 2: Read the value of **n**.

Step 3: If **n = 0**, return 0.

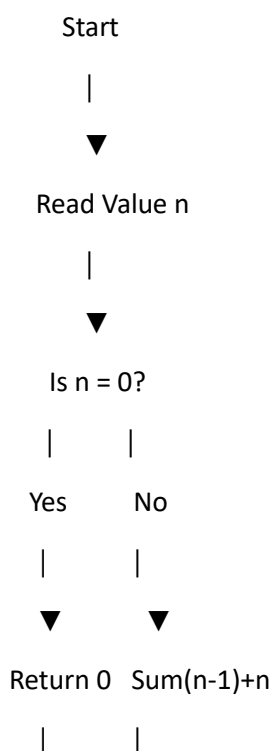
Step 4: Otherwise, recursively find the sum of **n-1**.

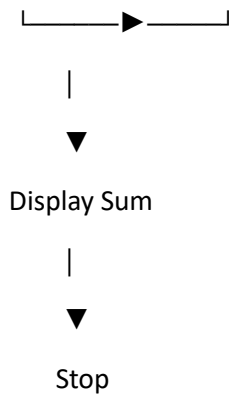
Step 5: Add **n** to the obtained sum.

Step 6: Display the result.

Step 7: Stop.

Flowchart





Prolog Program

sum(0, 0).

sum(N, S) :-

 N > 0,

 N1 is N - 1,

 sum(N1, S1),

 S is S1 + N.

Sample Output

?- sum(5, S).

S = 15.

Result

The Prolog program successfully calculated the sum of integers from **1 to n**.

Conclusion

Thus, the Prolog program was successfully implemented using recursion to compute the sum of natural numbers from **1 to n**.